

# Chapter 5

## Applications of Integration

### (積分的應用)

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- 5.1 Area of a Region Between Two Curves**
- 5.2 Volume: The Disk Method**
- 5.3 Volume: The Shell Method**
- 5.4 Arc Length and Surfaces of Revolution**



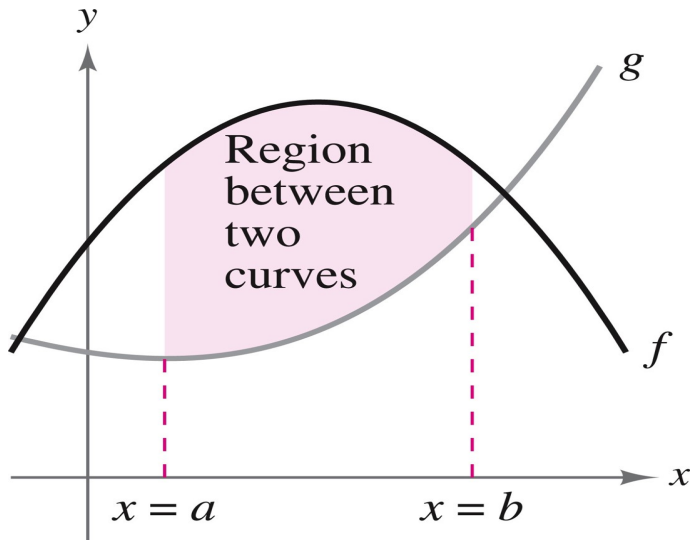
# Section 5.1

## Area of a Region Between Two Curves

### (兩曲線所夾區域的面積)

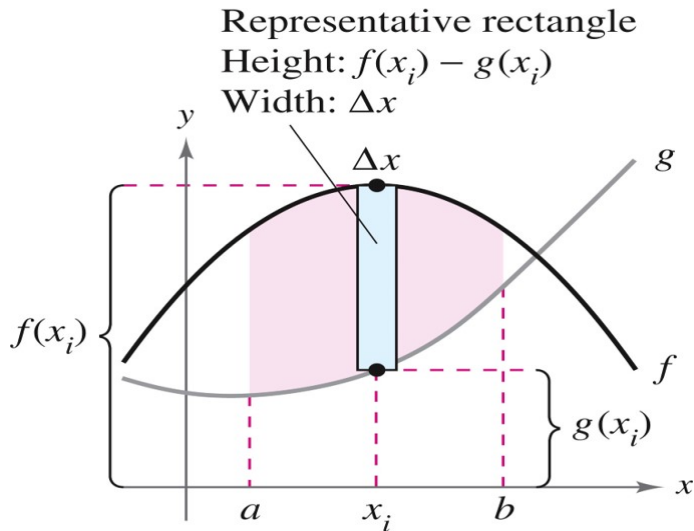


# Type I 的示意圖 (1/2)





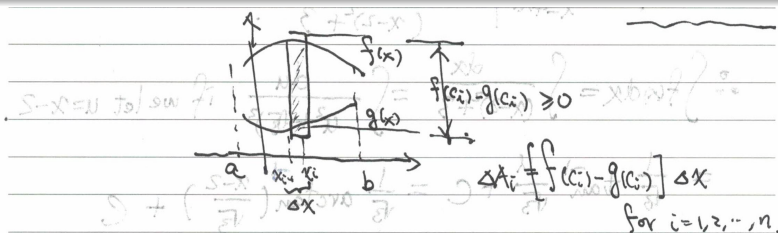
## Type I 的示意圖 (2/2)



## Type I (第一型面積公式)

If  $f$  and  $g$  are **conti.** on  $I = [a, b]$  with  $g(x) \leq f(x) \quad \forall x \in I$ , then the area of  $\mathcal{R} = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, g(x) \leq y \leq f(x)\}$  is given by

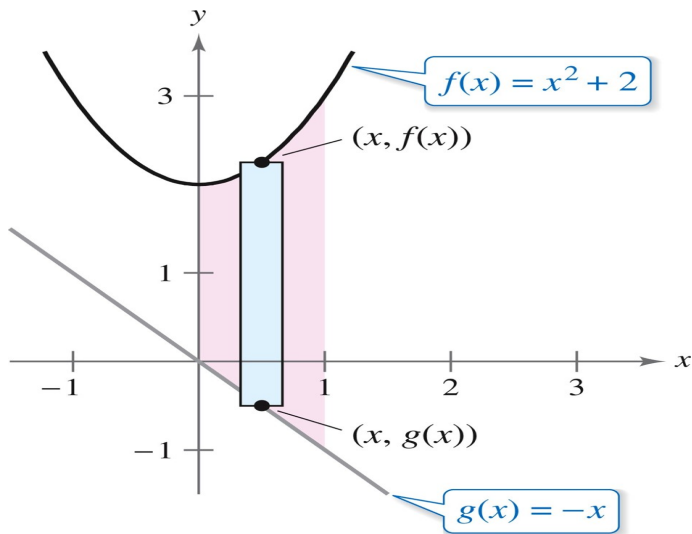
$$A = \text{area}(\mathcal{R}) = \lim_{n \rightarrow \infty} \sum_{i=1}^n [f(c_i) - g(c_i)] \Delta x = \int_a^b [f(x) - g(x)] dx \geq 0.$$

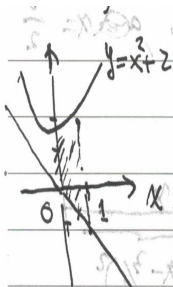


\* Example 1: Find the area of the region  $R$  bounded by the graphs of  $y = x^2 + 2$ ,  $y = -x$ ,  $x = 0$  and  $x = 1$ .



## Example 1 的示意圖 (承上頁)





Sol: Let  $g(x) = -x$  and  $f(x) = x^2 + 2 \quad \forall x \in [0, 1]$ .

$\Rightarrow g(x) \leq f(x) \quad \forall x \in [0, 1]$  and  $f$  and  $g$  are  
conti. on  $[0, 1]$ .

$$\Rightarrow A = \text{area}(R) = \int_0^1 [(x^2 + 2) - (-x)] dx$$

$$= \int_0^1 (x^2 + x + 2) dx = \left( \frac{1}{3}x^3 + \frac{1}{2}x^2 + 2x \right) \Big|_0^1 = \frac{17}{6}$$



### Remark (兩曲線所圍成的區域面積)

If  $f$  and  $g$  intersect (相交) at the points  $c_1, c_2, \dots, c_k$  for some  $k \in \mathbb{N}$ , then the area of the region between curves  $y = f(x)$  and  $y = g(x)$  is given by

$$\int_{c_1}^{c_2} |f(x) - g(x)| dx + \int_{c_2}^{c_3} |f(x) - g(x)| dx + \cdots + \int_{c_{k-1}}^{c_k} |f(x) - g(x)| dx.$$



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$$\int_a^b |f(x) - g(x)| dx$$

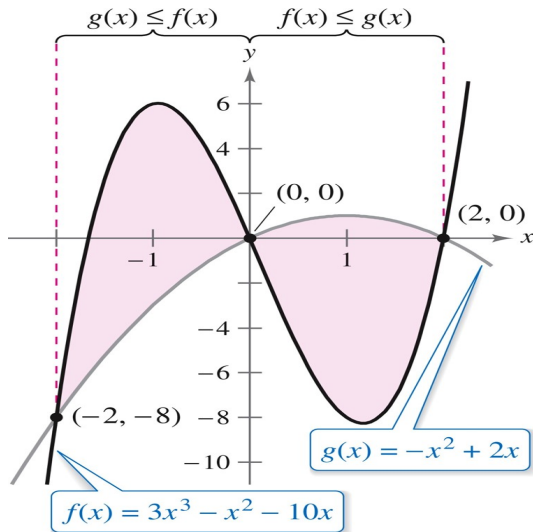
Example 4: (兩函數曲線有交點之情形)

Find the area of the region between the graphs

of  $f(x) = 3x^3 - x^2 - 10x$  and  $g(x) = -x^2 + 2x$ .



## Example 4 的示意圖 (承上頁)





Sol: Solve  $f(x)=g(x) \Leftrightarrow 3x^3 - x^2 - 10x = -x^2 + 2x$ .

$$\Leftrightarrow 3x^3 - 12x = 3x(x^2 - 4) = 3x(x-2)(x+2) = 0.$$

$$\Leftrightarrow x = -2, 0, 2.$$

$\therefore g(x) \leq f(x) \forall x \in [-2, 0]$  and  $f(x) \leq g(x) \forall x \in [0, 2]$ .

$$\therefore A = \text{area}(R) = \int_{-2}^0 [f(x) - g(x)] dx + \int_0^2 [g(x) - f(x)] dx.$$

$$= \int_{-2}^0 (3x^3 - 12x) dx + \int_0^2 (12x - 3x^3) dx.$$

$$= \left( \frac{3}{4}x^4 - 6x^2 \right) \Big|_{-2}^0 + \left( 6x^2 - \frac{3}{4}x^4 \right) \Big|_0^2 = \dots = \underline{\underline{24}}.$$



## Type II (第二型面積公式)

If  $f$  and  $g$  are **conti.** on  $I = [c, d]$  with  $g(y) \leq f(y) \quad \forall y \in I$ , then the area of the region between the graphs of  $x = f(y)$  and  $x = g(y)$

$$\mathcal{R} = \{(x, y) \in \mathbb{R}^2 \mid c \leq y \leq d, g(y) \leq x \leq f(y)\}$$

is given by

$$A = \text{area}(\mathcal{R}) = \int_c^d [f(y) - g(y)] dy \geq 0.$$



Example 7 : (Type I of 1343)

Find the area of the region between the graphs of  
 $x = f(y) = 3 - y^2$  and  $x = g(y) = y + 1$ .



Sol: Need to solve  $f(y) = g(y) \Leftrightarrow 3 - y^2 = y + 1$

$$\Leftrightarrow y^2 + y - 2 = (y+2)(y-1) = 0 \Leftrightarrow y = -2, 1$$

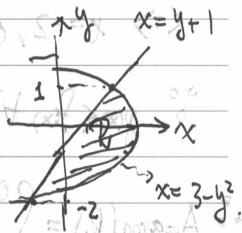
∵  $f(y)$  and  $g(y)$  are conti. on  $[-2, 1]$  and

$$f(y) \geq g(y) \quad \forall y \in [-2, 1].$$

$$\therefore A = \text{area}(R) = \int_{-2}^1 [f(y) - g(y)] dy$$

$$= \int_{-2}^1 (3 - y^2 - y - 1) dy = \int_{-2}^1 (2 - y^2 - y) dy$$

$$= \left( 2y - \frac{1}{3}y^3 - \frac{1}{2}y^2 \right) \Big|_{-2}^1 = \frac{9}{2}$$



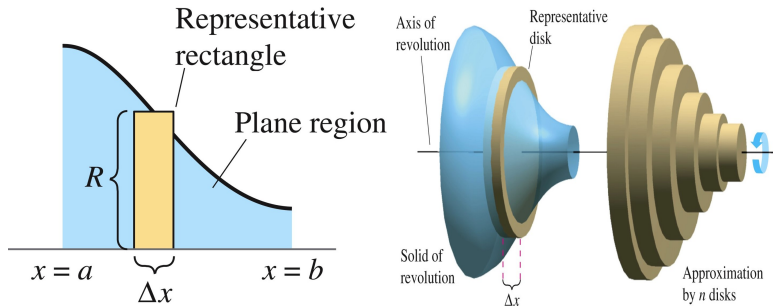
# Section 5.2

## Volume: The Disk Method

### (體積：圓盤法)

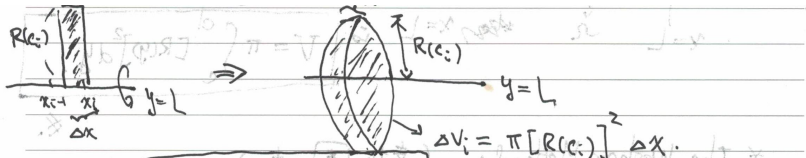


# A Solid of Revolution (旋轉體)



# The Volume of a Representative Disk

- Given a partition  $\{x_0, x_1, \dots, x_n\}$  of  $[a, b]$  with equal width  $\Delta x = \frac{b-a}{n}$ , where  $x_0 = a$  and  $x_n = b$ .
- For each  $i = 1, 2, \dots, n$ , choose  $c_i \in [x_{i-1}, x_i]$ , the volume of a disk of width  $\Delta x$  is  $\Delta V_i = \pi[R(c_i)]^2 \Delta x$ .



## Type I: Horizontal Axis of Revolution (水平旋轉軸)

Given the plane region  $\Omega$  and a horizontal axis of revolution  $y = L$ . If the radius function  $R(x) \geq 0$  is conti. on  $[a, b]$ , then the volume of a solid formed by revolving  $\Omega$  about  $y = L$  is

$$V = \lim_{n \rightarrow \infty} \sum_{i=1}^n \pi [R(c_i)]^2 \Delta x = \int_a^b \pi [R(x)]^2 dx \geq 0.$$





Example 1: Let  $\Omega$  be the plane region defined by

$$\Omega = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq \pi, 0 \leq y \leq \sqrt{\sin x}\}$$

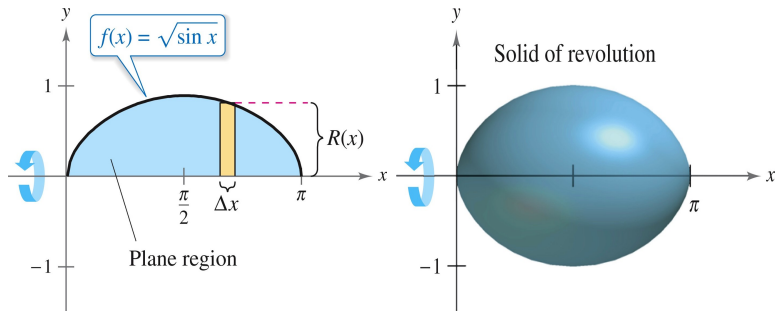
Find the volume of the solid formed by revolving  $\Omega$   
about the  $x$ -axis.



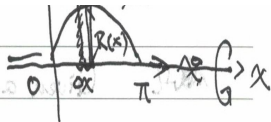
$$y = \sqrt{\sin x}$$



# Example 1 的示意圖 (承上頁)



Sol: Since the radius function



$R(x) = \sqrt{\sin x} \geq 0$  is conti. on  $[0, \pi]$ ,

$$V = \pi \int_0^{\pi} [R(x)]^2 dx = \pi \int_0^{\pi} \sin x dx = \pi (-\cos x) \Big|_0^{\pi}$$

$$= \underline{2\pi}$$



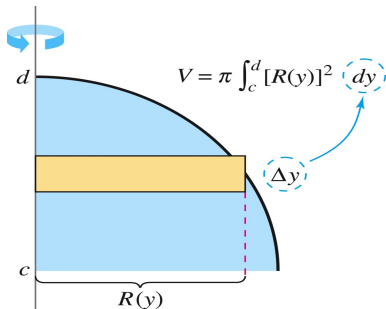
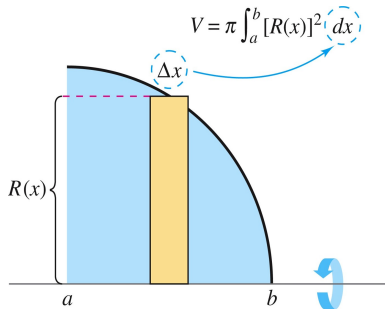
## Type II: Vertical Axis of Revolution (垂直旋轉軸)

Given the plane region  $\Omega$  and a vertical axis of revolution  $x = L$ . If the radius function  $R(y) \geq 0$  is conti. on  $[c, d]$ , then the volume of a solid formed by revolving  $\Omega$  about  $x = L$  is

$$V = \int_c^d \pi [R(y)]^2 dy \geq 0.$$

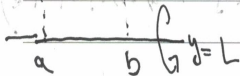


# 示意圖 (承上頁)



\* The Washer Method: (墊圓法)

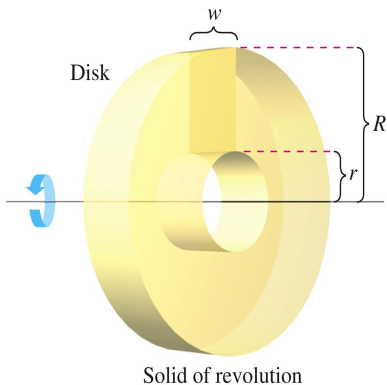
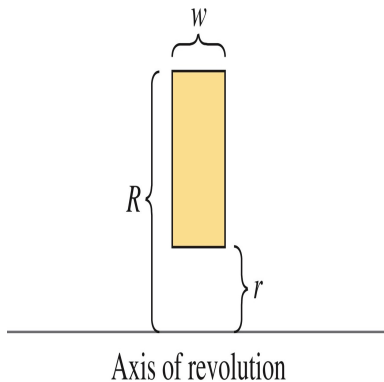
$\Omega$ : plane region



[Q]: What is the volume of a solid formed by revolving  $\Omega$  about  $y=L$ ?



# 墊圈 (washer) 的示意圖



# The Volume of a Representative Washer

- Given a partition  $\{x_0, x_1, \dots, x_n\}$  of  $[a, b]$  with equal width  $\Delta x = \frac{b-a}{n}$ , where  $x_0 = a$  and  $x_n = b$ .
- For each  $i = 1, 2, \dots, n$ , choose  $c_i \in [x_{i-1}, x_i]$ , the volume of a washer of width  $\Delta x$  is

$$\begin{aligned}\Delta V_i &= \pi[R(c_i)]^2 \Delta x - \pi[r(c_i)]^2 \Delta x \\ &= \pi \left( [R(c_i)]^2 - [r(c_i)]^2 \right) \Delta x.\end{aligned}$$





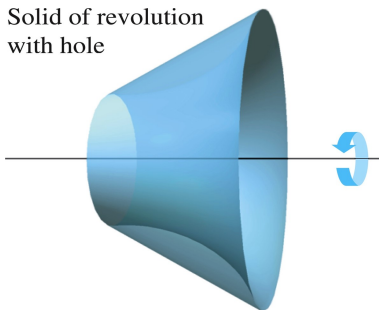
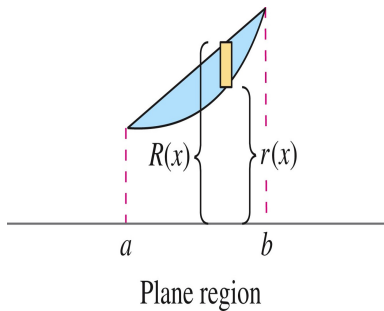
## Type I: Horizontal Axis of Revolution $y = L$

If the outer radius  $R(x) \geq 0$  and inner radius  $r(x) \geq 0$  are conti. on  $[a, b]$ , then the volume of a solid formed by revolving  $\Omega$  about  $y = L$  is

$$\begin{aligned} V &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \pi \left( [R(c_i)]^2 - [r(c_i)]^2 \right) \Delta x \\ &= \pi \int_a^b \left( [R(x)]^2 - [r(x)]^2 \right) dx. \end{aligned}$$



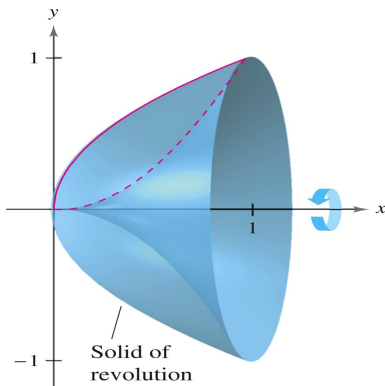
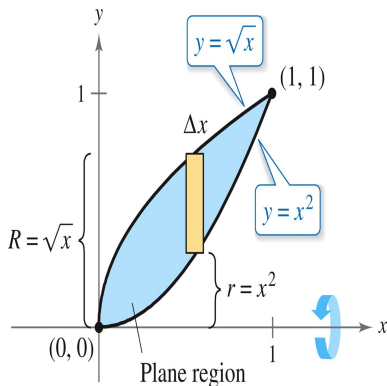
# 示意圖 (承上頁)

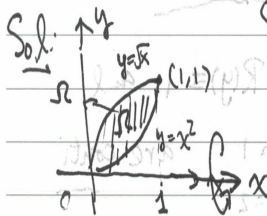


Example 3: Find the volume of the solid formed by revolving the region  $\Omega$  bounded by the graphs of  ~~$y = \sqrt{x}$~~   $y = \sqrt{x}$  and  $y = x^2$  about the  $x$ -axis.



# Example 3 的示意圖 (承上頁)





Sol: Solve  $y = \sqrt{x} = x^2$

$$\Rightarrow x = x^4 \Rightarrow x(x^3 - 1) = 0$$

$$\Rightarrow x = 0 \text{ or } x = 1.$$

Note that  $R(x) = \sqrt{x} \geq 0$  and  $r(x) = x^2 \geq 0$

are conti. on  $[0, 1]$ . So,

$$V = \pi \int_0^1 [(\sqrt{x})^2 - (x^2)^2] dx = \pi \int_0^1 (x - x^4) dx$$

$$= \pi \left( \frac{1}{2} x^2 - \frac{1}{5} x^5 \right) \Big|_0^1 = \frac{3}{10} \pi$$



## Type II: Vertical Axis of Revolution $x = L$

If the outer radius  $R(y) \geq 0$  and inner radius  $r(y) \geq 0$  are conti. on  $[c, d]$ , then the volume of a solid formed by revolving  $\Omega$  about  $x = L$  is

$$V = \pi \int_c^d \left( [R(y)]^2 - [r(y)]^2 \right) dy.$$



- Example 4 : (Type II of [3.1.3])

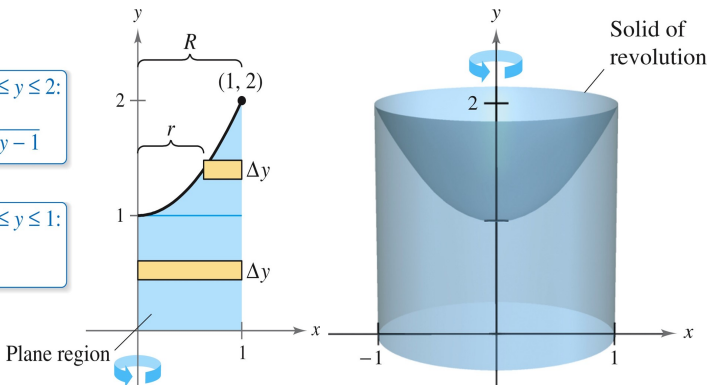
Find the volume of the solid formed by revolving  
the region  $\Sigma$  bounded by the graphs of  $y = x^2 + 1$ ,  
 $y = 0$ ,  $x = 0$  and  $x = 1$  about the  $y$ -axis.



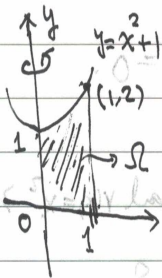
## Example 4 的示意圖 (承上頁)

For  $1 \leq y \leq 2$ :  
 $R = 1$   
 $r = \sqrt{y-1}$

For  $0 \leq y \leq 1$ :  
 $R = 1$   
 $r = 0$







Sol: Note that  $R(y) \equiv 1$  and

$$r(y) = \begin{cases} 0, & 0 \leq y \leq 1 \\ \sqrt{y-1}, & 1 \leq y \leq 2 \end{cases} \quad \text{are conti.}$$

on  $[0, 2]$ . Then

$$V = \int_0^1 (1^2 - 0^2) dy + \int_1^2 [1^2 - (\sqrt{y-1})^2] dy.$$

$$= \cancel{1} \pi + \pi \int_1^2 (2-y) dy = \pi + \pi \left( 2y - \frac{y^2}{2} \right) \Big|_1^2$$

$$= \pi + \pi \left( 2 - 2 + \frac{1}{2} \right) = \pi + \frac{\pi}{2} = \frac{3}{2} \pi$$



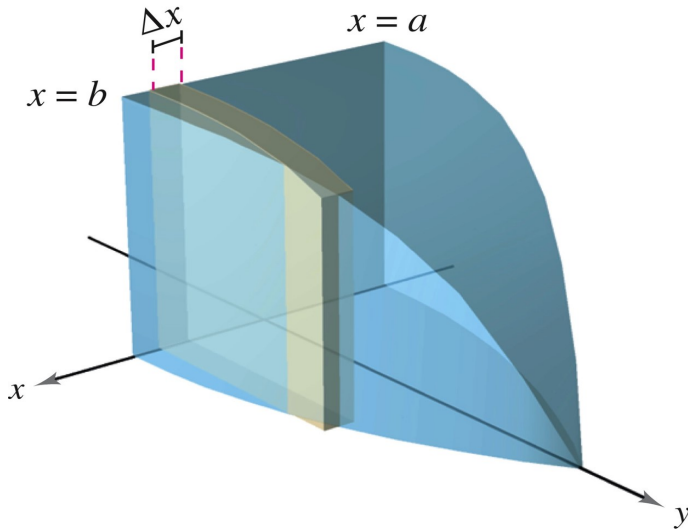
## Def (已知截面積的固體體積)

Let  $A(x)$  be the cross-sectional area (截面積) of a solid taken perpendicular to (垂直於) the  $x$ -axis at each  $x \in [a, b]$ . If  $A(x)$  is integrable on  $[a, b]$ , then the volume of the solid is

$$V = \int_a^b A(x) dx \geq 0.$$



# 示意圖 (承上頁)

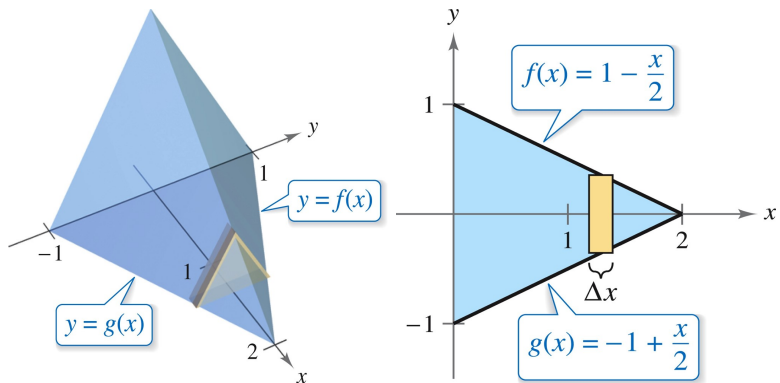


Example 6: The base of the solid is the region bounded by the graphs of  $f(x) = 1 - \frac{x}{2}$ ,  $g(x) = \sqrt{1 + \frac{x}{2}}$  and  $x = 0$ .

Find the volume of the solid if ~~the~~<sup>its</sup> cross sections perpendicular to the  $x$ -axis are equilateral triangles.  
(正三角形)

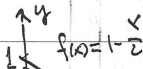


## Example 6 的示意圖 (承上頁)



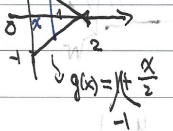
Sol:

the graphs of  $f$  and  $g$  intersect at  $(2, 0)$



A graph of the function  $f(x) = 1 - \frac{x}{2}$  on a Cartesian coordinate system. The line starts at  $(0, 1)$  and ends at  $(2, 0)$ . The x-axis is labeled with 0, 1, and 2. The y-axis is labeled with 1 and -1. A blue vertical line is drawn at  $x=1$ , and a blue horizontal line is drawn at  $y=0$ .

Since  $f(x) = g(x) \Leftrightarrow 1 - \frac{x}{2} = \frac{x}{2} \Leftrightarrow x = 2$ .



A graph of the function  $g(x) = \frac{x}{2}$  on a Cartesian coordinate system. The line starts at  $(0, 0)$  and ends at  $(2, 1)$ . The x-axis is labeled with 0, 1, and 2. The y-axis is labeled with 1 and -1. A blue vertical line is drawn at  $x=1$ , and a blue horizontal line is drawn at  $y=0$ .

For each  $x \in [0, 2]$ , the area of

the cross section is  $A(x) = \frac{\sqrt{3}}{4} (\text{base})^2$

$$= \frac{\sqrt{3}}{4} [f(x) - g(x)]^2 = \frac{\sqrt{3}}{4} (2-x)^2$$

So, the volume of the solid is

$$\begin{aligned} V &= \int_0^2 A(x) dx = \int_0^2 \frac{\sqrt{3}}{4} (2-x)^2 dx = \frac{\sqrt{3}}{12} (2-x)^3 \Big|_0^2 \\ &= 0 + \frac{\sqrt{3}}{12} \times 8 = \frac{2\sqrt{3}}{3} \end{aligned}$$



# Section 5.3

## Volume: The Shell Method

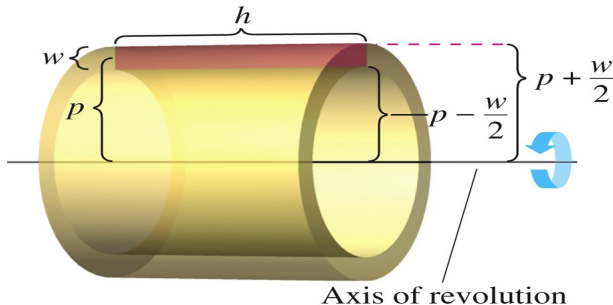
### (體積：殼層法)



# The Volume of a Shell

The volume of a shell of revolution is given by

$$\Delta V = \pi \left(p + \frac{w}{2}\right)^2 h - \pi \left(p - \frac{w}{2}\right)^2 h = 2\pi p h w.$$





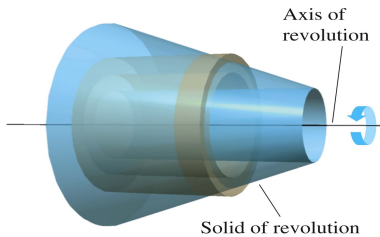
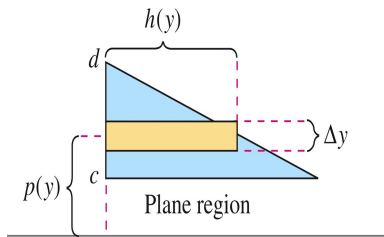
## Type I: Horizontal Axis of Revolution $y = L$

If  $p(y)$  and  $h(y)$  are conti. functions of  $y$  on  $[c, d]$ , then the volume of the solid formed by revolving a plane region  $\Omega$  about  $y = L$  is

$$V = \lim_{n \rightarrow \infty} \sum_{i=1}^n 2\pi p(c_i) h(c_i) \Delta y = 2\pi \int_c^d p(y) h(y) dy.$$



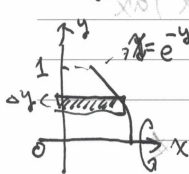
# 示意圖 (承上頁)



Example 2: Find the volume of the solid formed

by revolving  $R = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq y \leq 1, 0 \leq x \leq e^{-y^2}\}$

about the  $x$ -axis.



Sol:  $\because p(y) = y$  and  $h(y) = x = e^{-y^2}$   
are conti. on  $[0, 1]$ .

$$\therefore V = \int_0^1 2\pi p(y) h(y) dy = 2\pi \int_0^1 y e^{-y^2} dy.$$

$$= (2\pi) \left( -\frac{1}{2} \right) \int_0^1 e^{-y^2} (-2y) dy = (-\pi) e^{-y^2} \Big|_0^1$$

$$= (-\pi)(e^{-1} - 1) = \underline{\underline{\pi(1 - e^{-1}) \div 1.986}}$$



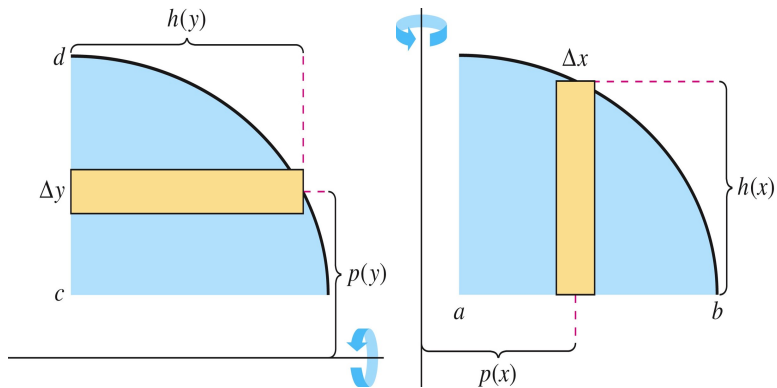
## Type II: Vertical Axis of Revolution $x = L$

If  $p(x)$  and  $h(x)$  are conti. functions of  $x$  on  $[a, b]$ , then the volume of the solid formed by revolving a plane region  $\Omega$  about  $x = L$  is

$$V = 2\pi \int_a^b p(x)h(x) dx.$$



# 如何選取函數 $p(\cdot)$ 和 $h(\cdot)$ ?

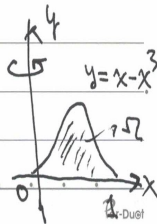


Example 1: (Washer Method 失敗不可行的例子)

Find the volume of the solid formed by revolving

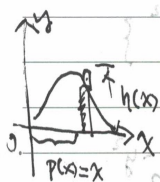
$$\Omega = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, 0 \leq y \leq x - x^3\}$$

about the  $y$ -axis.



Sol: Note that the washer Method fails for this example!

$\because p(x) = x$  and  $h(x) = y = x - x^3$  are conti. on  $[0, 1]$ .



$$\begin{aligned}
 \because V &= 2\pi \int_0^1 x(x - x^3) dx = 2\pi \int_0^1 (x^2 - x^4) dx \\
 &= 2\pi \left( \frac{1}{3} x^3 - \frac{1}{5} x^5 \right) \Big|_0^1 = 2\pi \left( \frac{1}{3} - \frac{1}{5} \right) = \underline{\underline{\frac{4\pi}{15}}}
 \end{aligned}$$

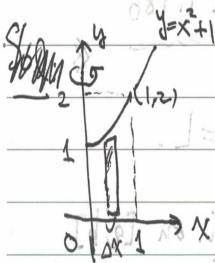


Example 3: (与 Example 4, Sec. 5.2 作比较)

Let  $\Omega = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, 0 \leq y \leq x^2 + 1\}$ . Find the volume of the solid formed by revolving  $\Omega$  about the  $y$ -axis.







Sol: Applying the Shell Method, we consider  $p(x) = x$  and  $h(x) = x^2 + 1 \quad \forall x \in [0, 1]$ .

$$\Rightarrow V = 2\pi \int_0^1 p(x)h(x)dx = 2\pi \int_0^1 x(x^2 + 1)dx$$

$$= 2\pi \int_0^1 (x^3 + x)dx = 2\pi \left( \frac{1}{4}x^4 + \frac{1}{2}x^2 \right) \Big|_0^1 = \frac{3\pi}{2}$$

Note: 此法比 Washer Method 更簡潔



# Section 5.4

## Arc Length and Surfaces of Revolution

### (弧長與旋轉體曲面)



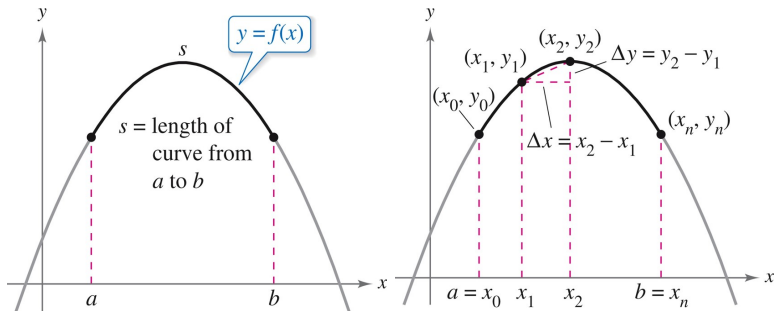
## Def (平滑曲線的定義)

Let  $f$  be a real-valued function defined on  $[a, b]$ .

- (1)  $f$  is continuously differentiable (連續可微) if its first derivative  $f'$  is conti. on  $[a, b]$ . In this case, we denote  $f \in C^1[a, b]$ .
- (2) The graph of  $f \in C^1[a, b]$  is called a smooth curve (平滑曲線).



# 弧長的示意圖



# The $i$ th Arc Length

For a smooth curve  $y = f(x)$  with  $f \in C^1[a, b]$ , the arc length of  $f$  on the subinterval  $[x_{i-1}, x_i]$  is given by

$$\Delta s_i = \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2} = \sqrt{1 + \left(\frac{\Delta y_i}{\Delta x_i}\right)^2} (\Delta x_i)$$

for each  $i = 1, 2, \dots, n$ . Moreover, it follows from M.V.T. that  $\exists c_i \in (x_{i-1}, x_i)$  s.t.  $f'(c_i) = \frac{\Delta y_i}{\Delta x_i}$ . Thus, we see that

$$\Delta s_i = \sqrt{1 + \left(\frac{\Delta y_i}{\Delta x_i}\right)^2} (\Delta x_i) = \sqrt{1 + [f'(c_i)]^2} (\Delta x_i)$$

for each  $i = 1, 2, \dots, n$ .



## Type I: 第一型弧長公式

The arc length of a smooth curve  $y = f(x)$  on  $[a, b]$  is

$$\begin{aligned}s &= \lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n \Delta s_i = \lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n \sqrt{1 + [f'(c_i)]^2} (\Delta x_i) \\ &= \int_a^b \sqrt{1 + [f'(x)]^2} dx.\end{aligned}$$



Example 2: (Type I 3m 長)

Find the arc length of the curve  $y = f(x) = \frac{x^3}{6} + \frac{1}{2x}$   
on the interval  $[\frac{1}{2}, 2]$ .



Sol:  $f'(x) = \frac{1}{2}x^2 - \frac{1}{2x^2} = \frac{1}{2}\left(x^2 - \frac{1}{x^2}\right)$ .

$$\therefore \sqrt{1 + [f'(x)]^2} = \sqrt{1 + \frac{1}{4}\left(x^2 - \frac{1}{x^2}\right)^2} = \sqrt{\frac{1}{4}\left(x^4 + 2 + \frac{1}{x^4}\right)}$$

$$= \sqrt{\frac{1}{4}\left(x^2 + \frac{1}{x^2}\right)^2} = \frac{1}{2}\left(x^2 + \frac{1}{x^2}\right) \text{ for } x \neq 0.$$

$\Rightarrow$  the arc length of  $y=f(x)$  between  $a$  and  $b$  is

$$s = \int_{\frac{1}{2}}^2 \frac{1}{2}\left(x^2 + \frac{1}{x^2}\right) dx = \frac{1}{2}\left(\frac{1}{3}x^3 - \frac{1}{x}\right) \Big|_{\frac{1}{2}}^2 = \frac{33}{16}.$$





✱

### Example 4 (Type I $\int \sqrt{1 - \frac{x^2}{a^2}}$ )

Find the arc length of the graph of  $y = f(x) = \ln(\cos x)$  from  $x=0$  to  $x=\pi/4$ .

Sol:  $\because \sqrt{1 + [f'(x)]^2} = \sqrt{1 + \left(\frac{-\sin x}{\cos x}\right)^2} = \sqrt{1 + \tan^2 x} = \sqrt{\sec^2 x} = \sec x \quad \forall x \in [0, \frac{\pi}{4}]$ .

$$\therefore s = \int_0^{\pi/4} \sec x \, dx = \ln |\sec x + \tan x| \Big|_0^{\pi/4} = \ln(\sqrt{2} + 1) \doteq 0.881$$



## Type II: 第二型弧長公式

The arc length of a smooth curve  $x = g(y)$  on  $[c, d]$  is

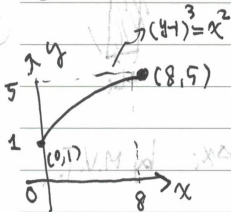
$$\begin{aligned}s &= \lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n \Delta s_i = \lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n \sqrt{1 + [g'(c_i)]^2} (\Delta y_i) \\ &= \int_c^d \sqrt{1 + [g'(y)]^2} dy.\end{aligned}$$



Example 3: (求 Type I 弧長).

Find the arc length of the curve  $(y-1)^3 = x^2$  on  $[0, 8]$ .

Sol: Consider  $x = g(y) = (y-1)^{3/2}$ .  $\forall y \in [1, 5]$ . Then



$$g'(y) = \frac{3}{2}(y-1)^{1/2} \text{ and hence}$$

$$\begin{aligned} \sqrt{1 + [g'(y)]^2} &= \sqrt{1 + \frac{9}{4}(y-1)} = \sqrt{\frac{9}{4}y - \frac{5}{4}} \\ &= \frac{1}{2} \sqrt{9y - 5}. \end{aligned}$$



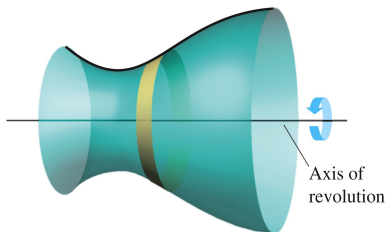
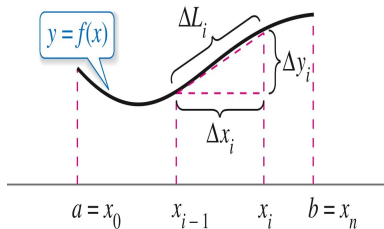
$S_0$ , the arc length is  $s = \int_1^5 \frac{1}{2} \sqrt{9y-5} dy$ .

$$= \frac{1}{18} \int_1^5 (9y-5)^{\frac{1}{2}} (9) dy = \frac{1}{18} \cdot \frac{2}{3} (9y-5)^{\frac{3}{2}} \Big|_1^5$$

$$= \frac{1}{27} (9y-5)^{\frac{3}{2}} \Big|_1^5 = \frac{1}{27} (40^{\frac{3}{2}} - 4^{\frac{3}{2}}) \doteq 9.073$$



# Area of Surfaces of Revolution (旋轉體的表面積)



## Main Question

What is the surface area  $S$  formed by revolving a smooth curve  $y = f(x)$  about the horizontal axis of revolution  $y = L$ ?

On the  $i$ th subinterval  $I_i = [x_{i-1}, x_i]$ , choose  $c_i, d_i \in I_i$ , we let

$r(d_i)$  = the radius of revolution at  $d_i \in I_i$ ,

$\Delta L_i$  = the arc length of the smooth curve on  $I_i$

$$= \sqrt{1 + [f'(c_i)]^2}(\Delta x_i),$$

where  $r(x) \geq 0$  is conti. on  $[a, b]$ .



### Note (第 $i$ 個旋轉體表面積)

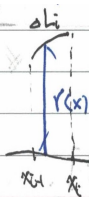
If  $d_i \in [x_{i-1}, x_i]$  ( $i = 1, 2, \dots, n$ ), it follows from M.V.T. that  $\exists c_i \in (x_{i-1}, x_i)$  s.t. the  $i$ th surface area of revolution is

$$\Delta S_i \approx 2\pi r(d_i) \cdot \Delta L_i = 2\pi r(d_i) \sqrt{1 + [f'(c_i)]^2} (\Delta x_i)$$

for  $i = 1, 2, \dots, n$ .

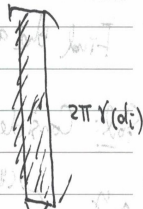


# 示意圖 (承上頁)



$r(d_i)$  = the radius of revolution at  $d_i \in (x_{i-1}, x_i)$ .

$\Delta L_i$  = the arc length of  $f$  on  $[x_{i-1}, x_i]$ .



the  $i$ th surface area of revolution is





## Type I: Horizontal Axis of Revolution $y = L$

If  $r(x) \geq 0$  is conti. on  $[a, b]$  and  $f \in C^1[a, b]$ , then the area of the surface formed by revolving a curve  $y = f(x)$  about  $y = L$  is

$$\begin{aligned} S &= \lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n 2\pi r(d_i) \sqrt{1 + [f'(c_i)]^2} (\Delta x_i) \\ &= 2\pi \int_a^b r(x) \sqrt{1 + [f'(x)]^2} dx. \end{aligned}$$



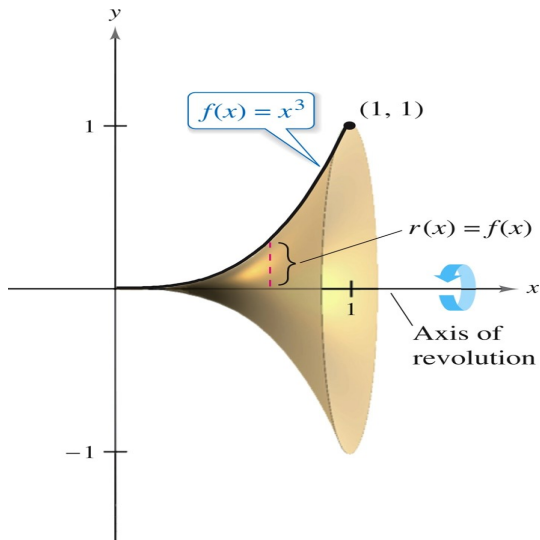
Example 6: (求 Type I 旋轉表面積).

✖

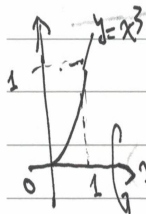
Find the area of surface formed by revolving the curve  $y=x^3$  about the  $x$ -axis ( $0 \leq x \leq 1$ ).



## Example 6 的示意圖 (承上頁)



Sol:  $\because r(x) = x^3$  and  $f'(x) = 3x^2 \quad \forall x \in [0, 1]$ .



$$\therefore S = \int_0^1 2\pi x^3 \sqrt{1 + (3x^2)^2} dx = 2\pi \int_0^1 x^3 \sqrt{1 + 9x^4} dx.$$

$$= \frac{2\pi}{36} \int_0^1 (36x^3)(1+9x^4)^{1/2} dx = \frac{\pi}{27} (1+9x^4)^{3/2} \Big|_0^1$$

$$= \frac{\pi}{27} (10^{3/2} - 1) \doteq 3.563.$$



## Type II: Vertical Axis of Revolution $x = L$

If  $r(y) \geq 0$  is conti. on  $[c, d]$  and  $g \in C^1[c, d]$ , then the area of the surface formed by revolving a curve  $x = g(y)$  about  $x = L$  is

$$\begin{aligned} S &= \lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n 2\pi r(d_i) \sqrt{1 + [g'(c_i)]^2} (\Delta y_i) \\ &= 2\pi \int_c^d r(y) \sqrt{1 + [g'(y)]^2} dy. \end{aligned}$$



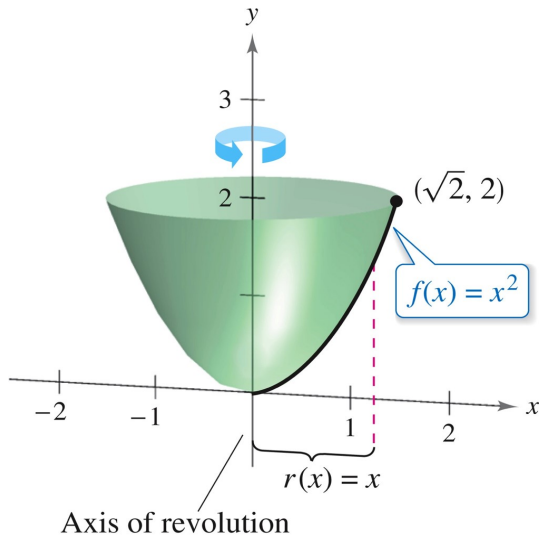
Example 7: (求 Type II 旋轉曲面面積)

Find the area of surface formed by revolving the curve  $y = x^2$  on

$[0, \sqrt{2}]$  about the  $y$ -axis.



## Example 7 的示意圖 (承上頁)



**Sol:** (使用教科書上的方法)

For the smooth curve  $y = f(x) = x^2$  on  $[0, \sqrt{2}]$ , consider the radius function  $r(x) = x$  and then we see that

$$\sqrt{1 + [f'(x)]^2} = \sqrt{1 + (2x)^2} = \sqrt{1 + 4x^2}.$$

So, the area of surface formed by revolving the curve  $y = f(x)$  about the  $y$ -axis is given by

$$\begin{aligned} S &= 2\pi \int_a^b r(x) \sqrt{1 + [f'(x)]^2} dx = 2\pi \int_0^{\sqrt{2}} x \sqrt{1 + 4x^2} dx \\ &= \frac{\pi}{4} \int_0^{\sqrt{2}} (8x)(1 + 4x^2)^{1/2} dx = \frac{\pi}{6} (1 + 4x^2)^{3/2} \Big|_0^{\sqrt{2}} = \frac{13\pi}{3}. \end{aligned}$$





**Sol:** (使用 Type II 的方法)

From  $y = x^2$ , we know that  $x = g(y) = y^{1/2} \quad \forall y \in [0, 2]$ . Hence, the radius function  $r(y) = g(y) = y^{1/2} \geq 0$  is conti. on  $[0, 2]$  and

$$\sqrt{1 + [g'(y)]^2} = \sqrt{1 + (\frac{1}{2}y^{-1/2})^2} = \sqrt{1 + \frac{1}{4}y^{-1}}.$$

So, the surface area of revolution about the  $y$ -axis is

$$\begin{aligned} S &= 2\pi \int_c^d r(y) \sqrt{1 + [g'(y)]^2} dy = 2\pi \int_0^2 y^{1/2} \sqrt{1 + \frac{1}{4}y^{-1}} dy \\ &= \pi \int_0^2 \sqrt{4y + 1} dy = \frac{\pi}{6} (4y + 1)^{3/2} \Big|_0^2 = \frac{13\pi}{3}. \end{aligned}$$



**Thank you for your attention!**

